

Bishwajit Saha

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RESEARCH SUMMARY

Research Scientist at IBM Research with first-author publications at **ICLR 2026**, **NeurIPS 2024 Workshop** and **ICML 2023**, and a submission under review at **NeurIPS 2026**. Research focus on energy-based language models, associative memory, Mixture-of-Experts, clustering and large-scale distributed training of billion-parameter LLMs. Four years of prior production-systems engineering at scale.

EDUCATION

Rensselaer Polytechnic Institute

Ph.D. in Computer Science

Troy, NY, USA

January, 2021 - January, 2025

Rensselaer Polytechnic Institute

M.Sc in Computer Science

Troy, NY, USA

January, 2021 - December, 2022

Bangladesh University of Engineering & Technology

B.Sc in Computer Science and Engineering

Dhaka, Bangladesh

February, 2013 - September, 2017

RESEARCH & PUBLICATIONS

Nima Dehmamy*, Benjamin Hoover*, **Bishwajit Saha***, Leo Kozachkov, Jean-Jacques Slotine and Dmitry Krotov*, “NRGPT: An Energy-based Alternative for GPT”, **ICLR 2026**. [Paper](#).

- o Proposed NRGPT, a language model architecture integrating associative memory dynamics that reconceptualizes GPT inference as a dynamical process on an energy landscape, demonstrating strong performance across diverse tasks from algebraic reasoning to open-domain text generation.

Bharat Runwal, **Bishwajit Saha**, Benjamin Hoover, Mauro Martino, Nima Dehmamy, “Boltzmann Routing for Energy-Compatible Mixture of Experts”, **NFAM Workshop @ICLR 2026**. [Paper](#).

- o Introduced Boltzmann Routing, an auxiliary-loss-free method that derives MoE expert selection from a free-energy functional, proving exact energy compatibility and a formal attention-routing duality within Energy Transformers.

Bharat Runwal, **Bishwajit Saha**, Benjamin Hoover, Dmitry Krotov, Mauro Martino, Nima Dehmamy. “Adaptive Test-Time Compute with Energy Transformer Modules.” **Under review at NeurIPS 2026**.

- o Identified two distinct recurrent-transformer computation modes through a 9-architecture factorial study at 400M–1B scale — distributed refinement (energy attention) vs. concentrated construction (standard attention) — enabling 12.8–19% adaptive-halting FLOP savings at no quality loss.

“Scaling NRGPT: Energy-based Language Models at Billion-Parameter Scale”, Work-in-progress, **IBM Research**, 2026.

- o Scaling the NRGPT (ICLR 2026) architecture to multi-billion parameters using distributed training on high-performance computing clusters, targeting competitive performance with GPT and other baseline language models while preserving the energy-based associative memory dynamics.

Saha, Bishwajit, Dmitry Krotov, Mohammed J. Zaki, and Parikshit Ram. “Deep Clustering with Associative Memories.” **Workshop @ICLR 2025 & Workshop @NeurIPS 2024**. [Paper](#).

- o Integrated autoencoder and associative memories together to perform end-to-end differentiable deep clustering (**DCIAM**) which simultaneously learns effective representations and clusters in the latent

space with a simplified deep clustering formulation and achieves significant improvement.

Saha, Bishwajit, Dmitry Krotov, Mohammed J. Zaki, and Parikshit Ram. “End-to-end Differentiable Clustering with Associative Memories.” **ICML 2023**. [Paper](#).

- o Uncovered a novel connection between the Associative Memories (AM) dynamics and the inherent discrete assignment necessary in clustering to propose a novel unconstrained continuous relaxation of the discrete clustering problem, enabling end-to-end differentiable clustering with AM (**CIAM**).

Rasul, Rownak Ara, Promy Saha, Diponkor Bala, S. M. Karim, Ibrahim Abdullah, and **Bishwajit Saha**. “An Evaluation of Machine Learning Approaches for Early Diagnosis of Autism Spectrum Disorder.” **Healthcare Analytics 2023**. [Paper](#).

Ibtehaz, Nabil, Shafayat Ahmed, **Bishwajit Saha**, M. Sohel Rahman, and Md Shamsuzzoha Bayzid. “NORTH: a highly accurate and scalable Naive Bayes based ORTHologous gene clustering algorithm.” *bioRxiv* (2019). [Paper](#).

WORK EXPERIENCE

Research Scientist

IBM Research

Cambridge, MA, USA

February 2025 - Present

Research Topic: Energy-based Language Models, Large-scale Distributed Training, Mixture-of-Experts.

Graduate Research Assistant

Rensselaer Polytechnic Institute

Troy, NY, USA

January 2021 - January 2025

Advisor: [Mohammed J. Zaki](#)

Research Topic: Clustering, Associative Memories, Machine Learning.

Summer AI Researcher (2 internships)

IBM Research

Cambridge, MA, USA

May 2022 & May 2023

Research Topic: End-to-end Differentiable Clustering with Associative Memories (**ICML 2023, NeurIPS 2024 Workshop**).

Senior Engineer

R&D, Technology Division, Bkash Ltd

Dhaka, Bangladesh

May 2019 - January 2021

- o Built loan origination system and payment gateway integrations (VISA, banks) for bKash, Bangladesh’s largest mobile financial platform (65M+ users). Stack: Java Spring Boot, Nodejs, PostgreSQL, AWS, Flutter.

Senior Software Engineer

REVE Systems Ltd

Dhaka, Bangladesh

October 2017 - April 2019

- o Developed government-scale web platforms (OISF, BCC Billing) using Java Spring Boot and MySQL.

Lecturer, Department of Computer Science and Engineering

Bangladesh University of Business and Technology

Dhaka, Bangladesh

August 2017 - September 2017

Courses Taught: Structured Programming Language, Data Structure, Graphics Design.

TECHNICAL SKILLS

Languages

Python, C, C++, Java, CUDA, Dart, Matlab, Prolog, Haskell, Scala

ML & DL

PyTorch, JAX, TensorFlow, Scikit-learn, FSDP

Infrastructure

Multi-node GPU Training, LSF/Slurm, HPC Clusters, AWS, Docker

Models & Methods

Transformers, Energy-based Models, Associative Memories, MoE, LoRA

PROFESSIONAL SERVICE & HONORS

- o **Reviewer**: NeurIPS 2023–2026, ICML 2024–2026, ICLR 2025–2026, AAAI 2026, AISTATS 2025.
- o Undergraduate Merit Position 25/150, Department of CSE, BUET. Govt. Board **Scholarship** 2007–2017.